

Road network elevation consideration

This document provides details about the consideration of the elevation of road segments in our project. Road network datasets were extracted from OpenStreetMap, and elevation values were obtained using the [Google Maps Elevation API](#). The API returns elevation in meters relative to the local mean sea level (LMSL), which we assigned to the nodes of each road network model.

To ensure that elevated road structures were accurately represented, we randomly sampled five networks from each of seven world regions and verified that these structures were correctly captured (**Figure 1**). In total, 35 networks distributed across 26 countries and regions were analyzed. The following pages present examples of selected networks that contain elevated road structures, along with the corresponding node elevation values.

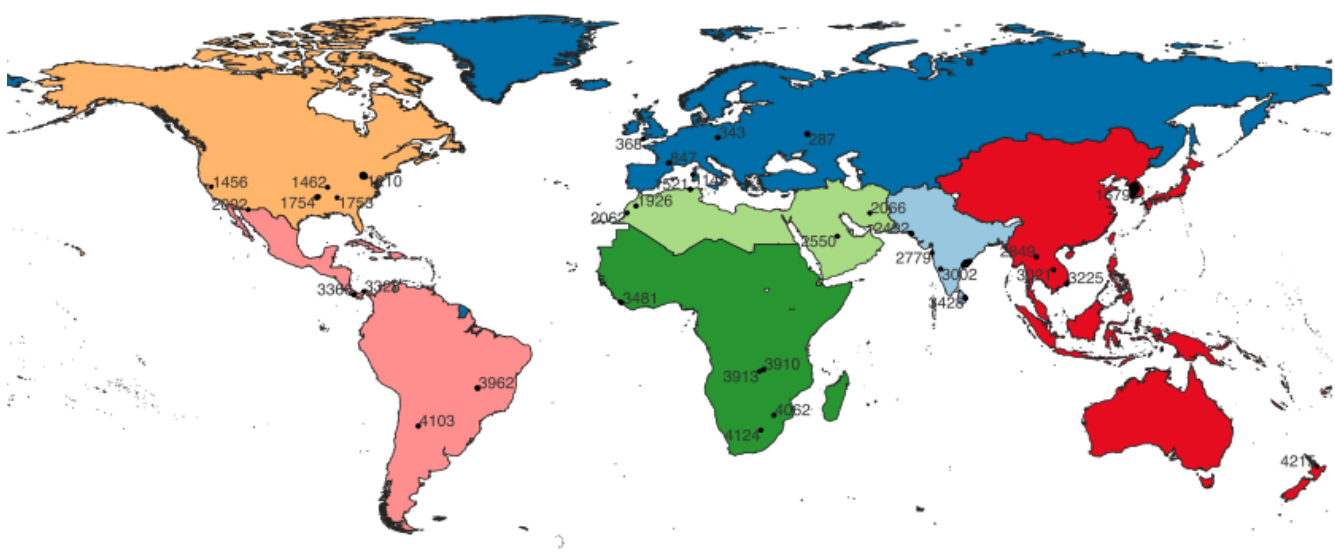


Figure 1. Sampled road networks in seven world regions. For each world region, we randomly sampled five settlement clusters and examine the accuracy of road segment elevation represented in the network model.

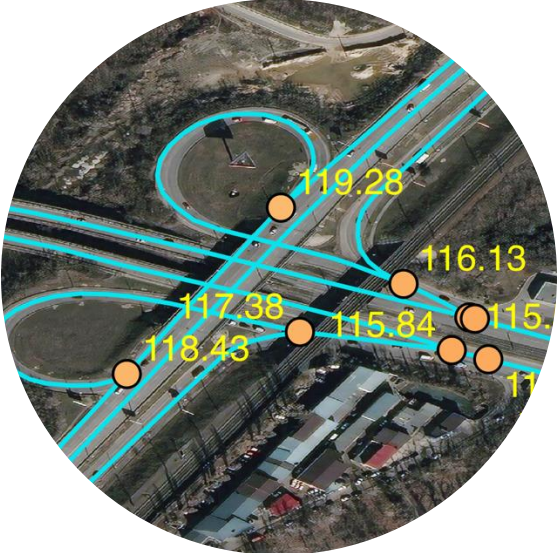
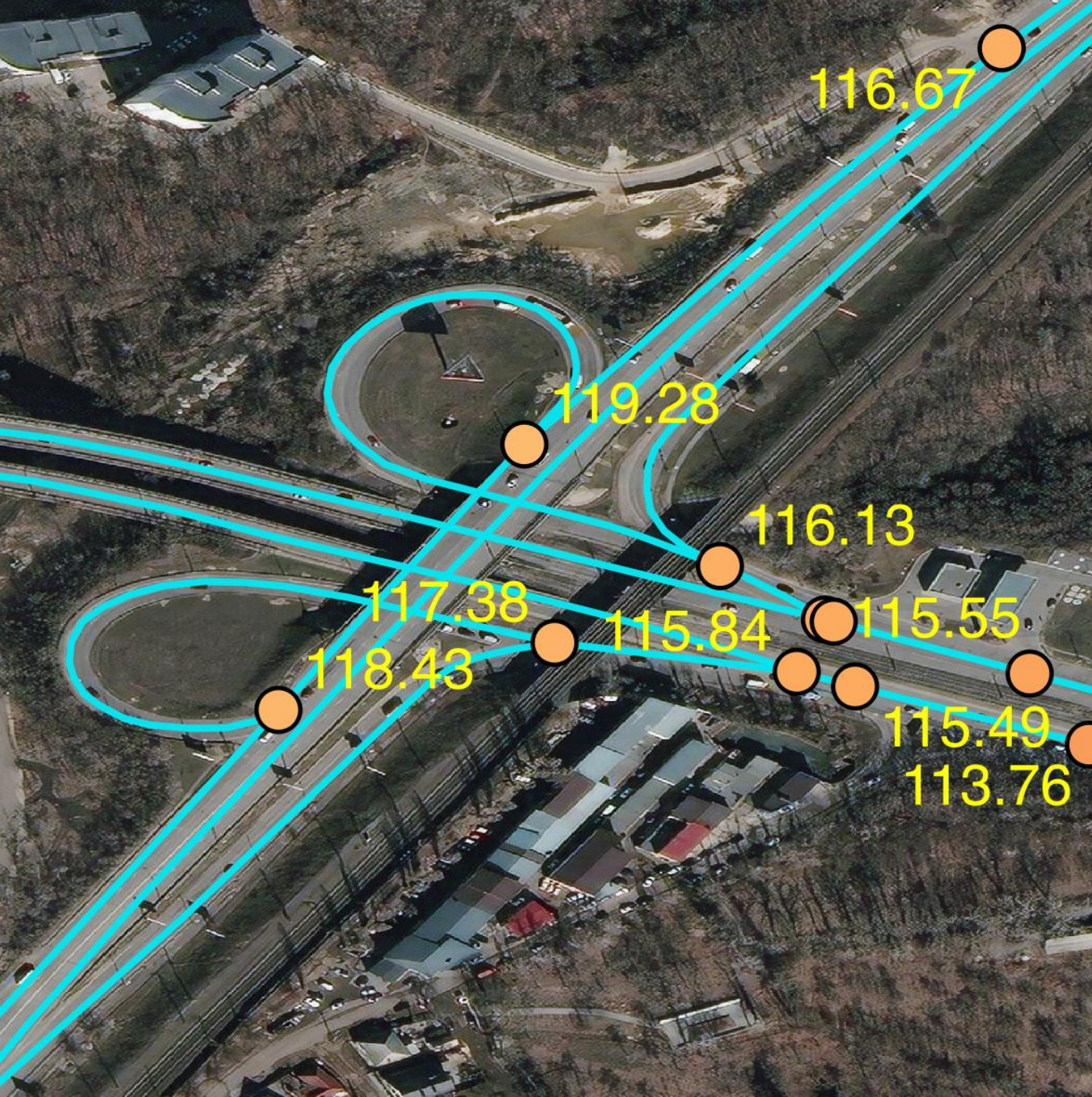
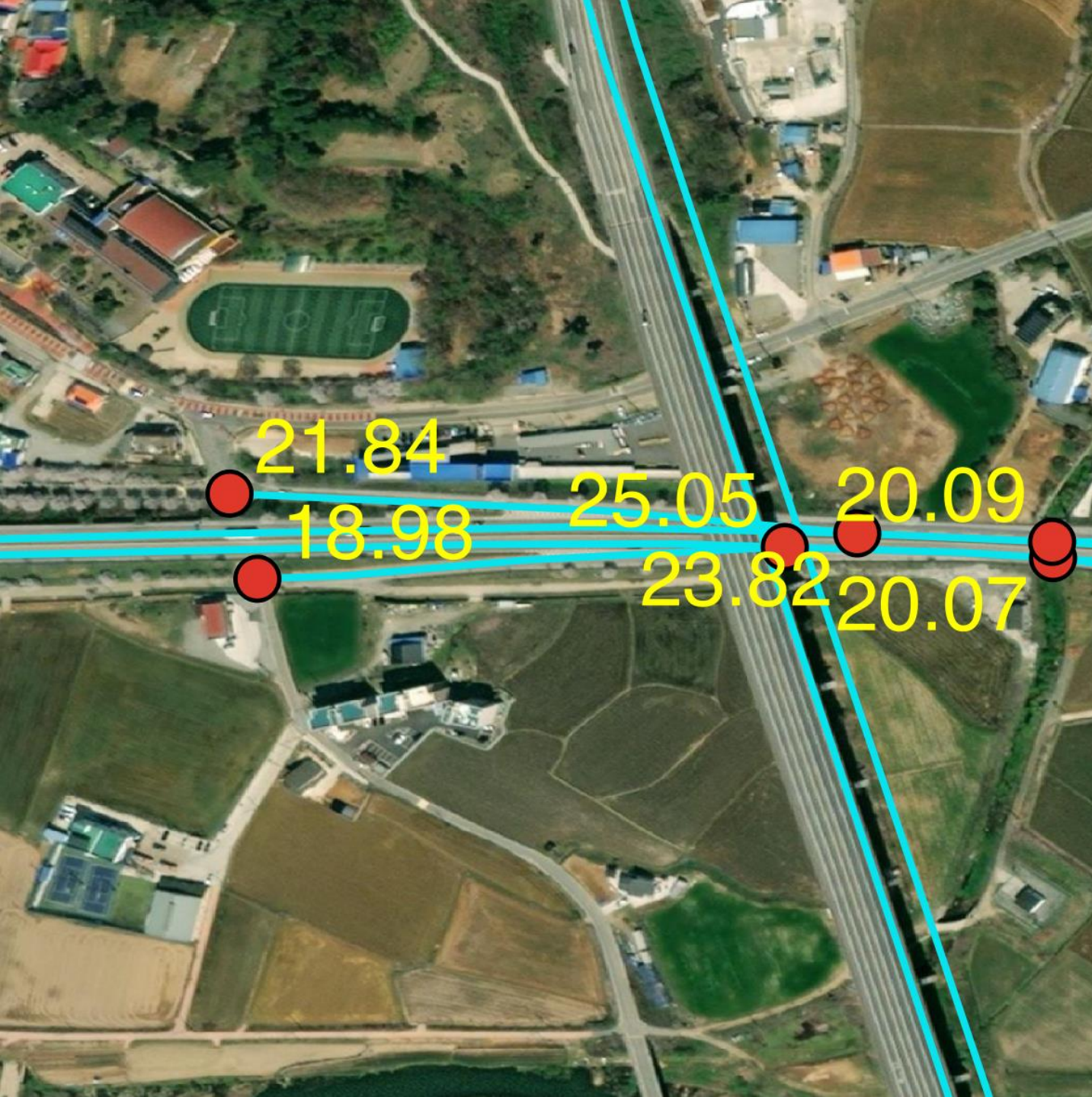


Figure 2. Representation of elevated road structures in the network model. Nodes are depicted as orange points, and road segments (edges) are shown as neon blue lines. Node elevations, expressed in meters relative to local mean sea level, are annotated in yellow. The basemap imagery is sourced from [Bing Satellite](#).



Network ID	287
Closest city	Voronezh
Country	Russia
World Region	Europe & Central Asia
Elevated road structure	Overpass
Difference in elevation	3.73 meters





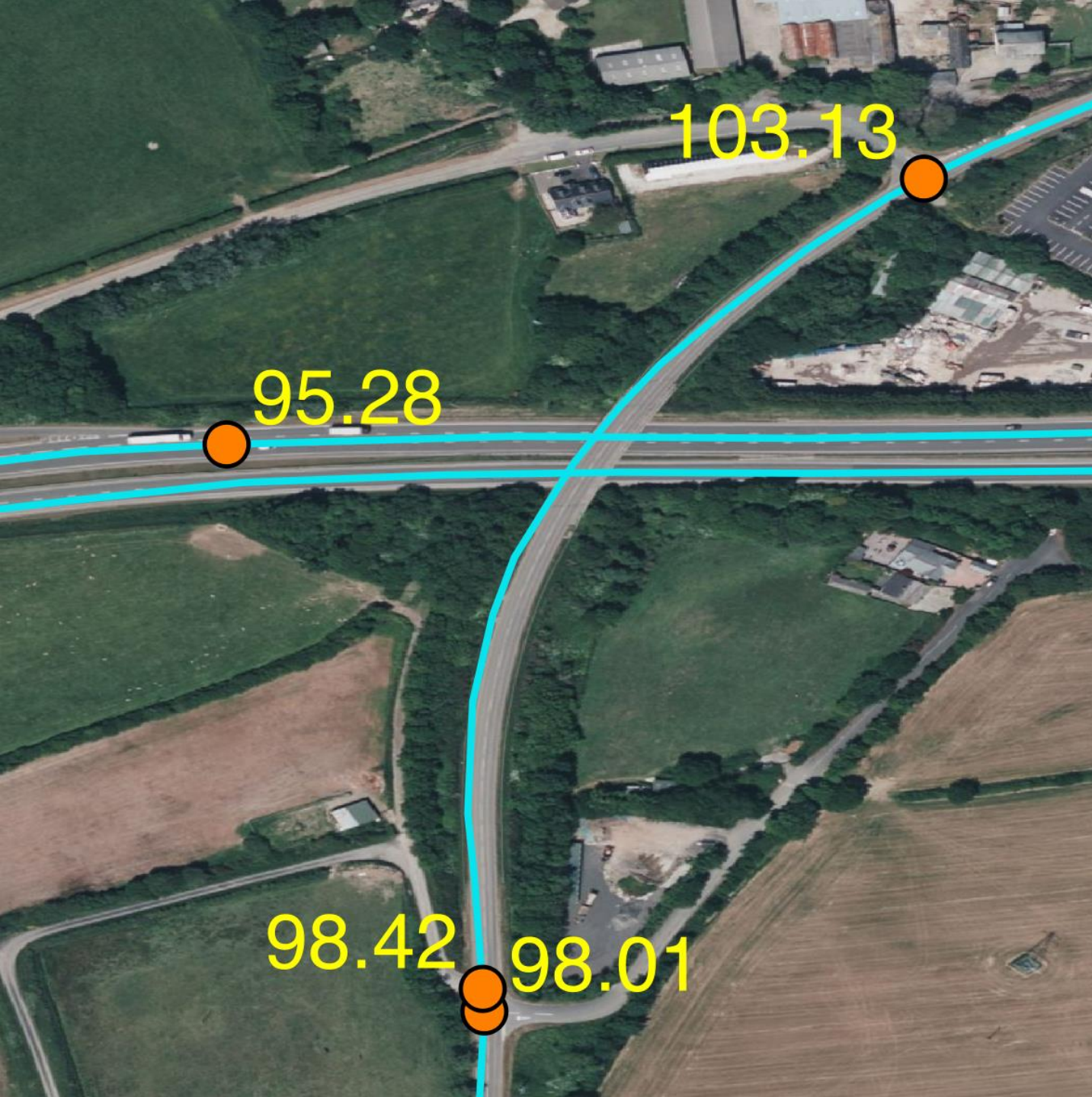
Network ID	1679
Closest city	Seoul
Country	Korea
World Region	East Asia & Pacific
Elevated road structure	Overpass
Difference in elevation	4.96 meters





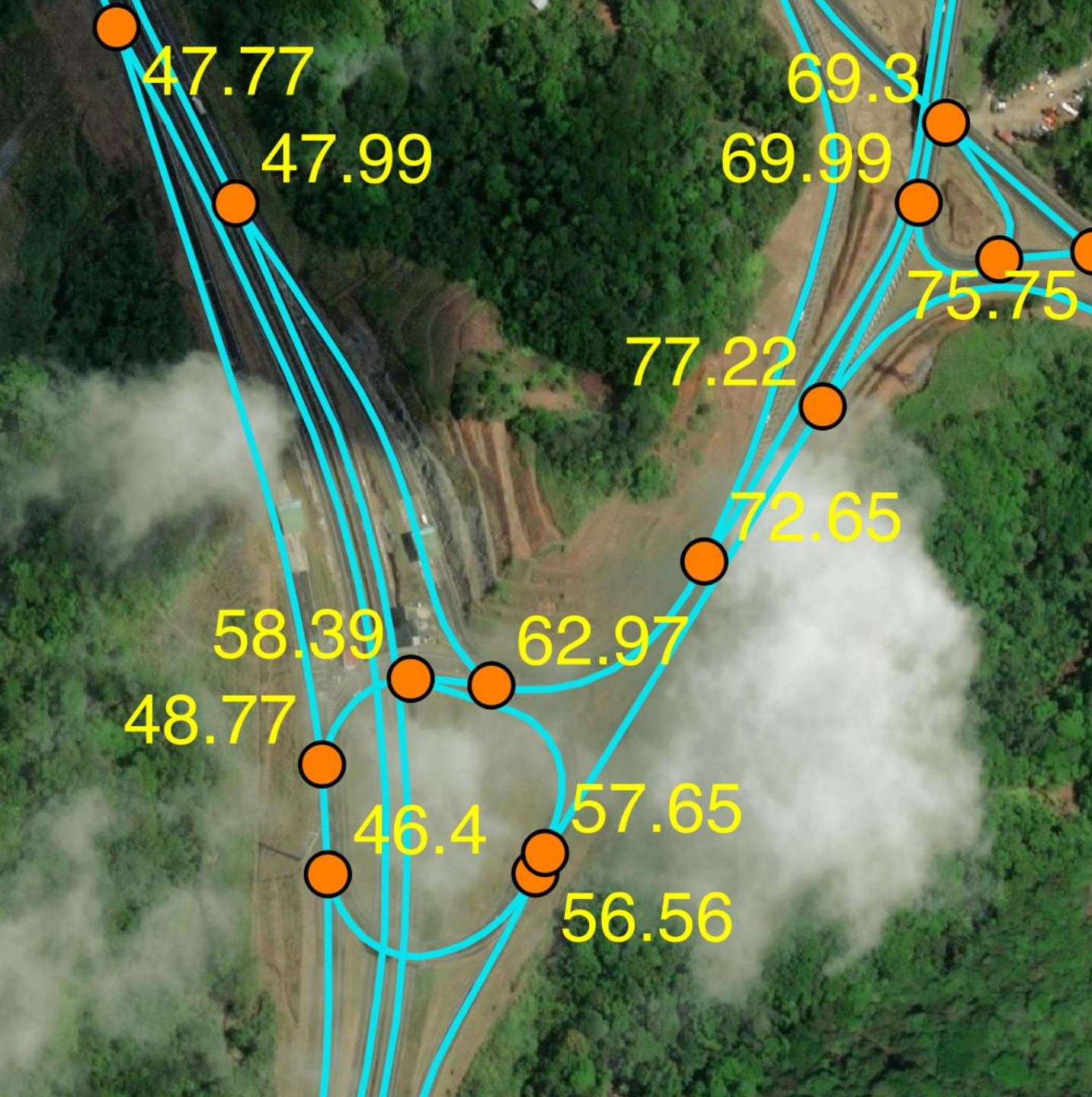
Network ID	1143
Closest city	Sassari
Country	Italy
World Region	Europe & Central Asia
Elevated road structure	Overpass
Difference in elevation	3.03 meters





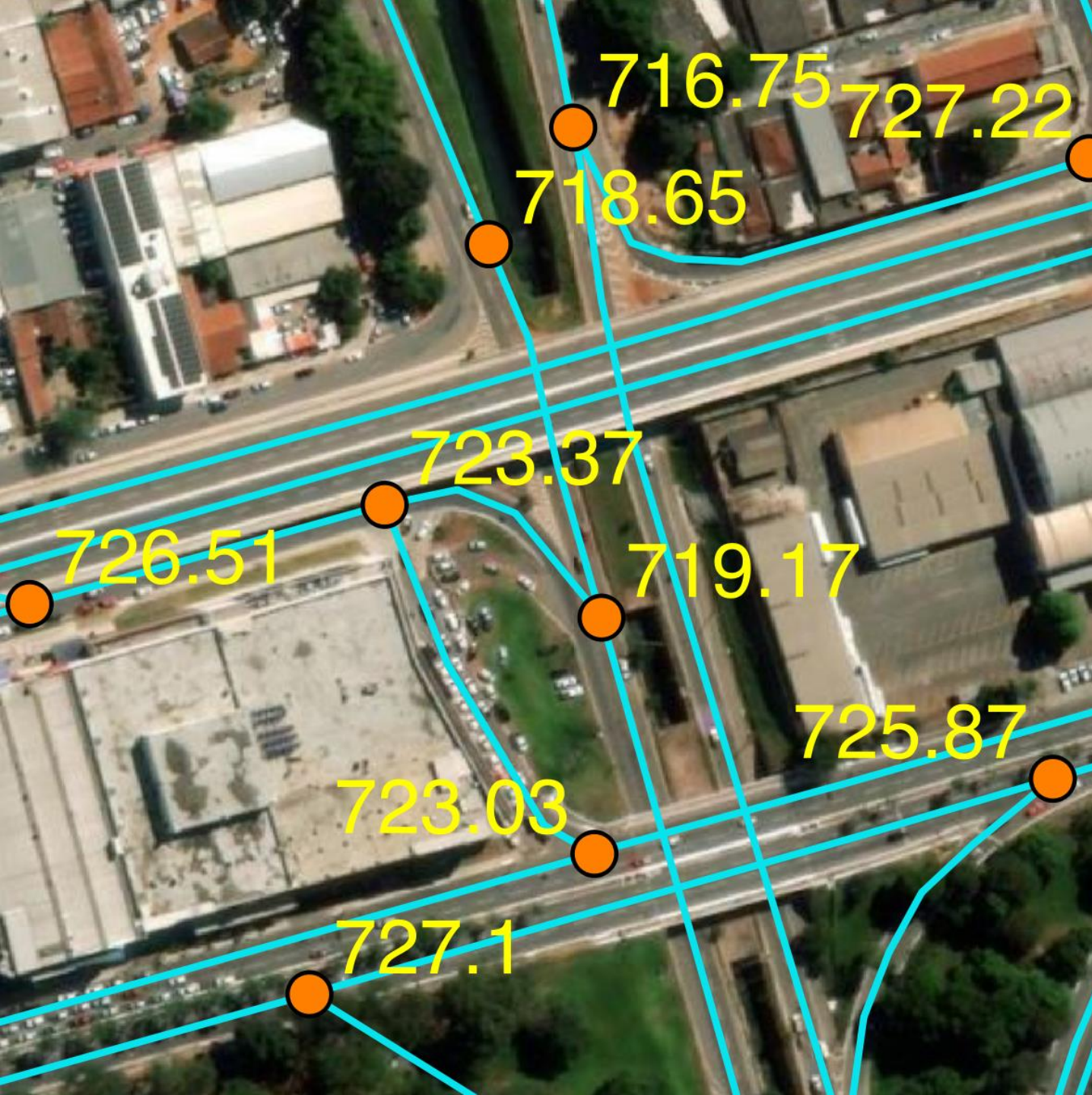
Network ID	368
Closest city	St Austell
Country	United Kingdom
World Region	Europe & Central Asia
Elevated road structure	Overpass
Difference in elevation	7.85 meters





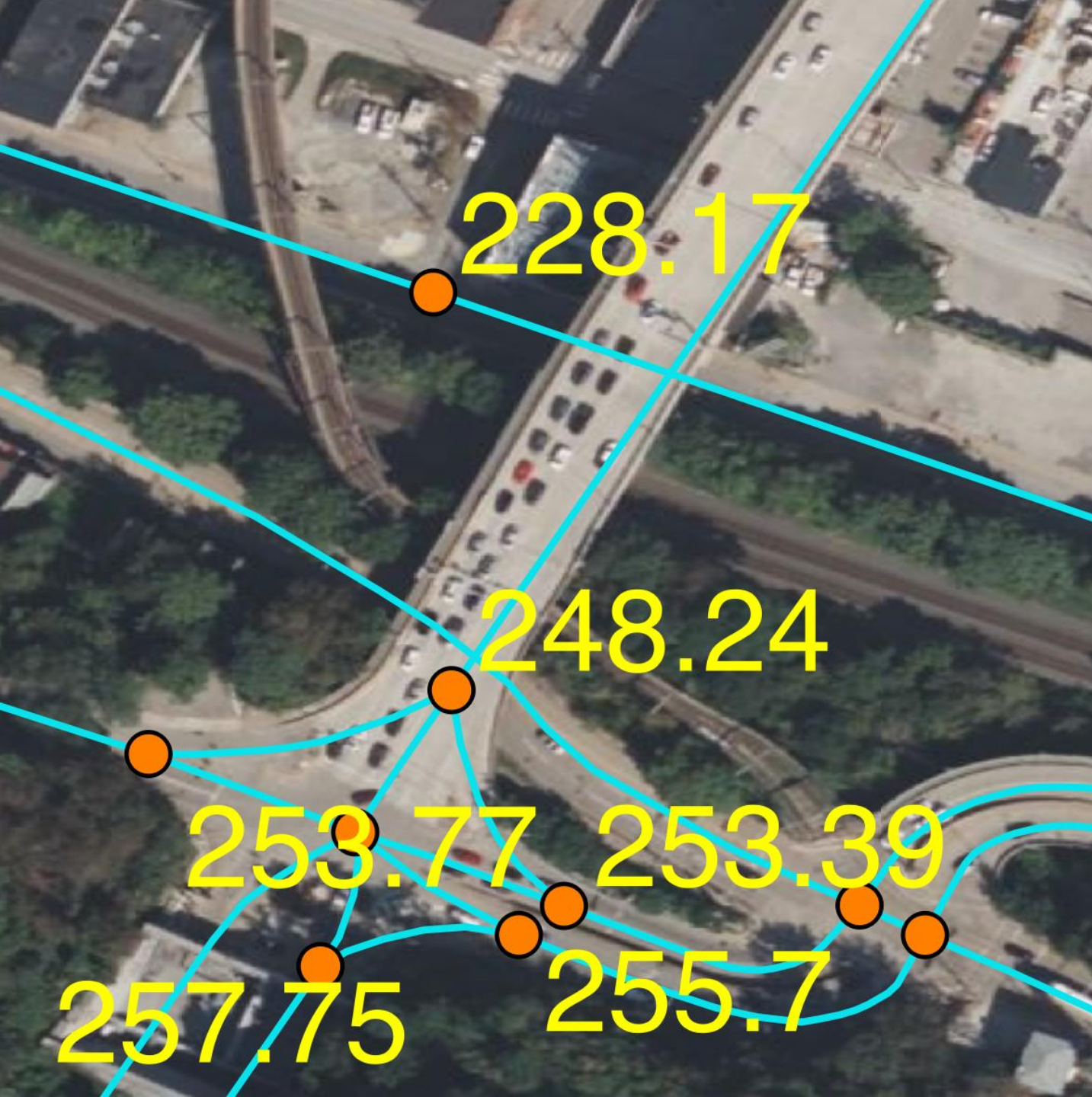
Network ID	3328
Closest city	Colón
Country	Panama
World Region	Latin America & Caribbean
Elevated road structure	Elevated roadway
Difference in elevation	10.4 meters





Network ID	3962
Closest city	St. Leste Vila Nova
Country	Brazil
World Region	Latin America & Caribbean
Elevated road structure	Overpass
Difference in elevation	8.44 meters





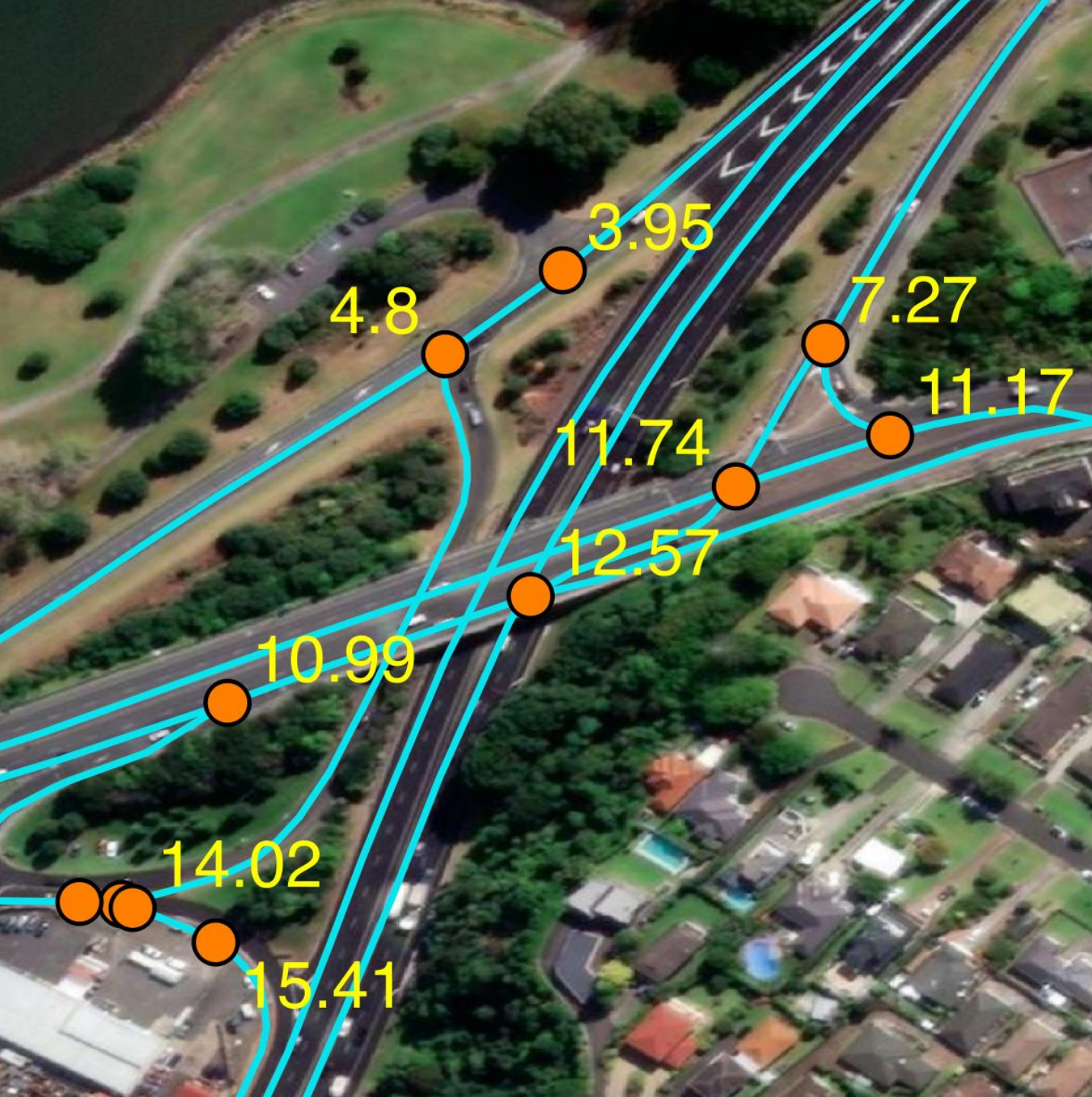
Network ID	1210
Closest city	Pittsburgh
Country	United States
World Region	North America
Elevated road structure	Overpass
Difference in elevation	20.07 meters





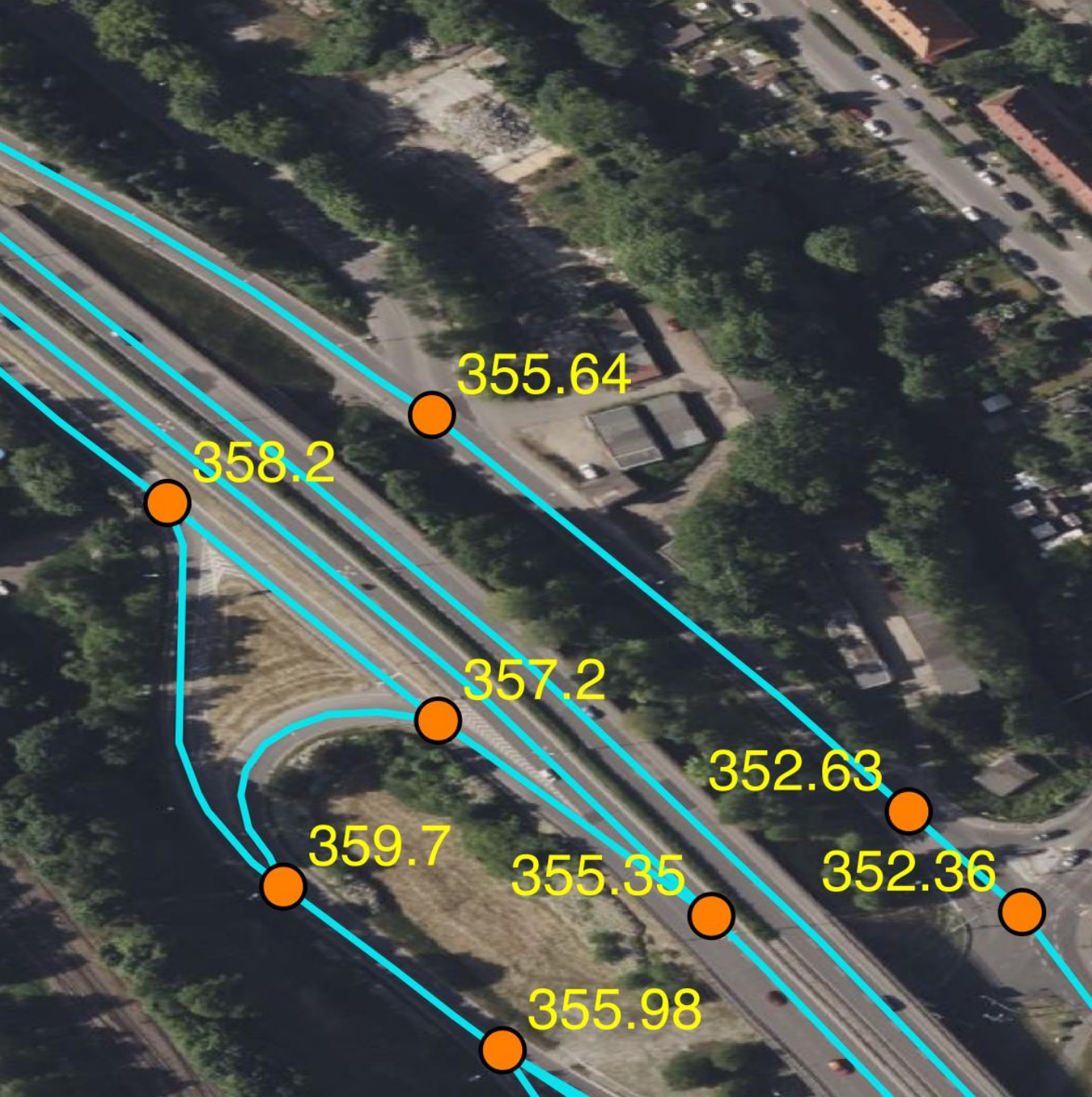
Network ID	1679
Closest city	Seoul
Country	Korea
World Region	East Asia & Pacific
Elevated road structure	Bridge
Difference in elevation	5.55 meters





Network ID	4215
Closest city	Tauranga
Country	New Zealand
World Region	East Asia & Pacific
Elevated road structure	Overpass
Difference in elevation	7.77 meters





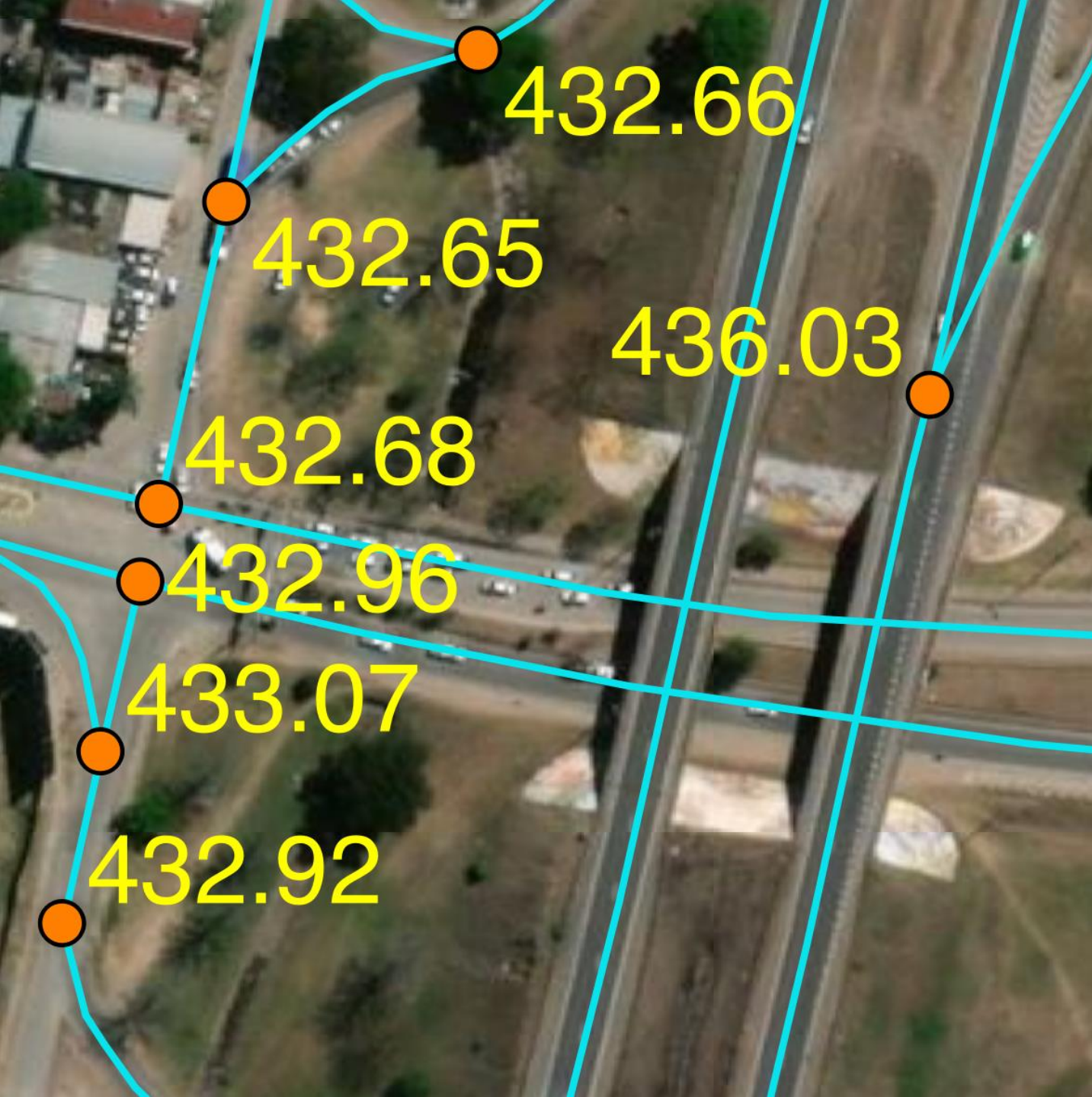
Network ID	343
Closest city	Jablonec nad Nisou
Country	Czechia
World Region	Europe & Central Asia
Elevated road structure	Elevated expressway
Difference in elevation	5.84 meters





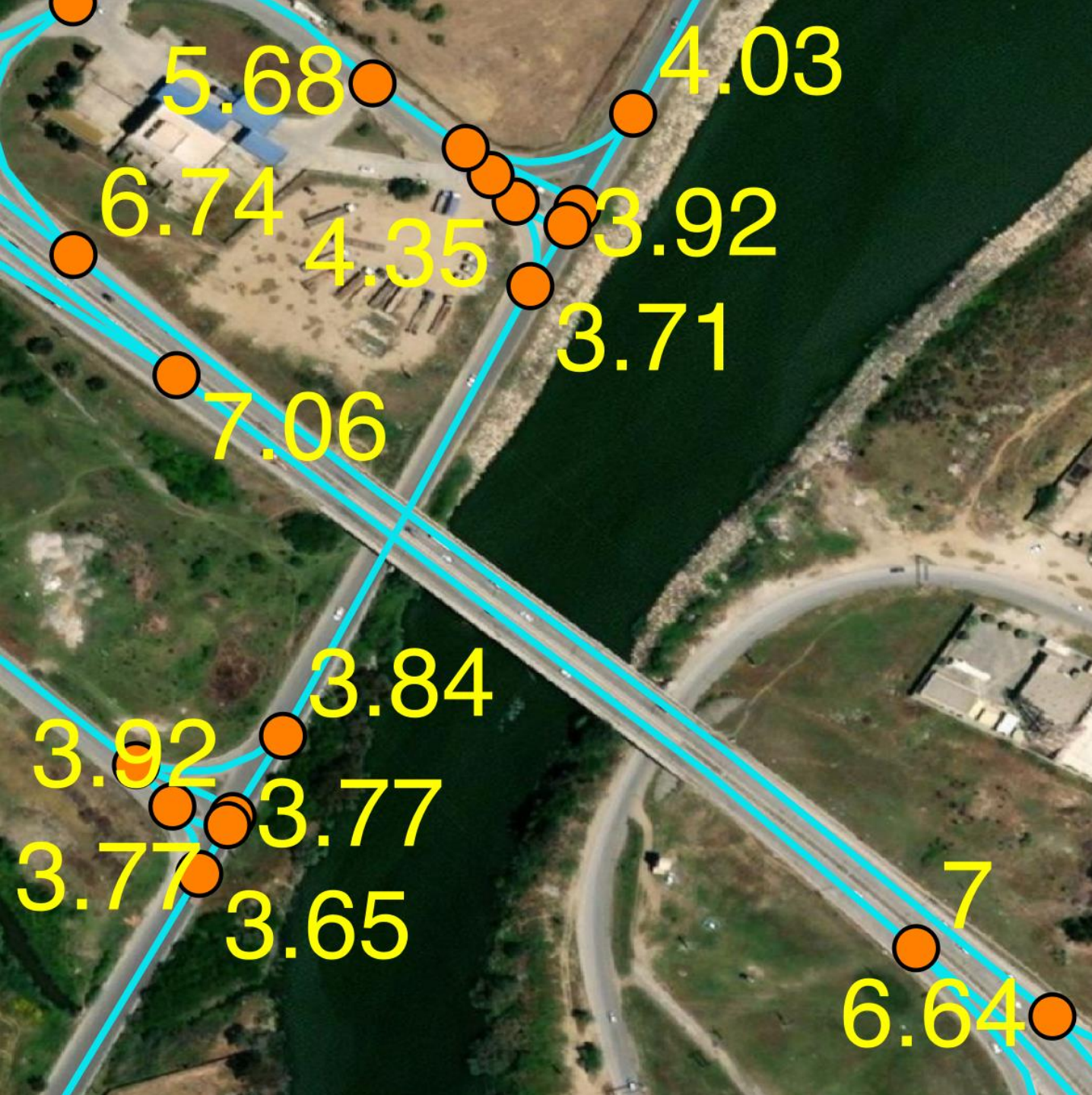
Network ID	2002
Closest city	Nogales
Country	Mexico
World Region	Latin America & Caribbean
Elevated road structure	Overpass
Difference in elevation	6.23 meters





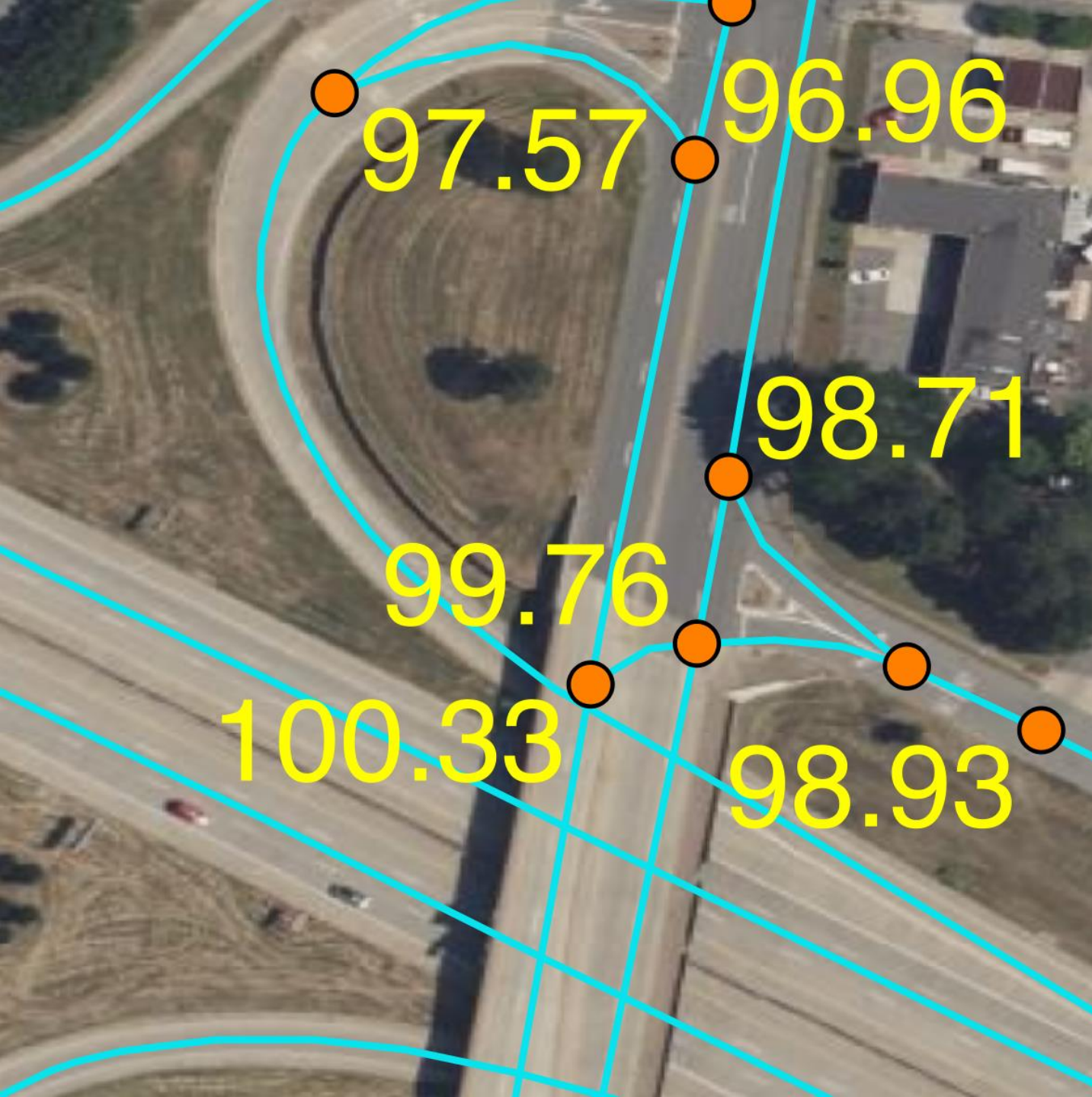
Network ID	4103
Closest city	San Miguel de Tucumán
Country	Argentina
World Region	Latin America & Caribbean
Elevated road structure	Overpass
Difference in elevation	3.35 meters





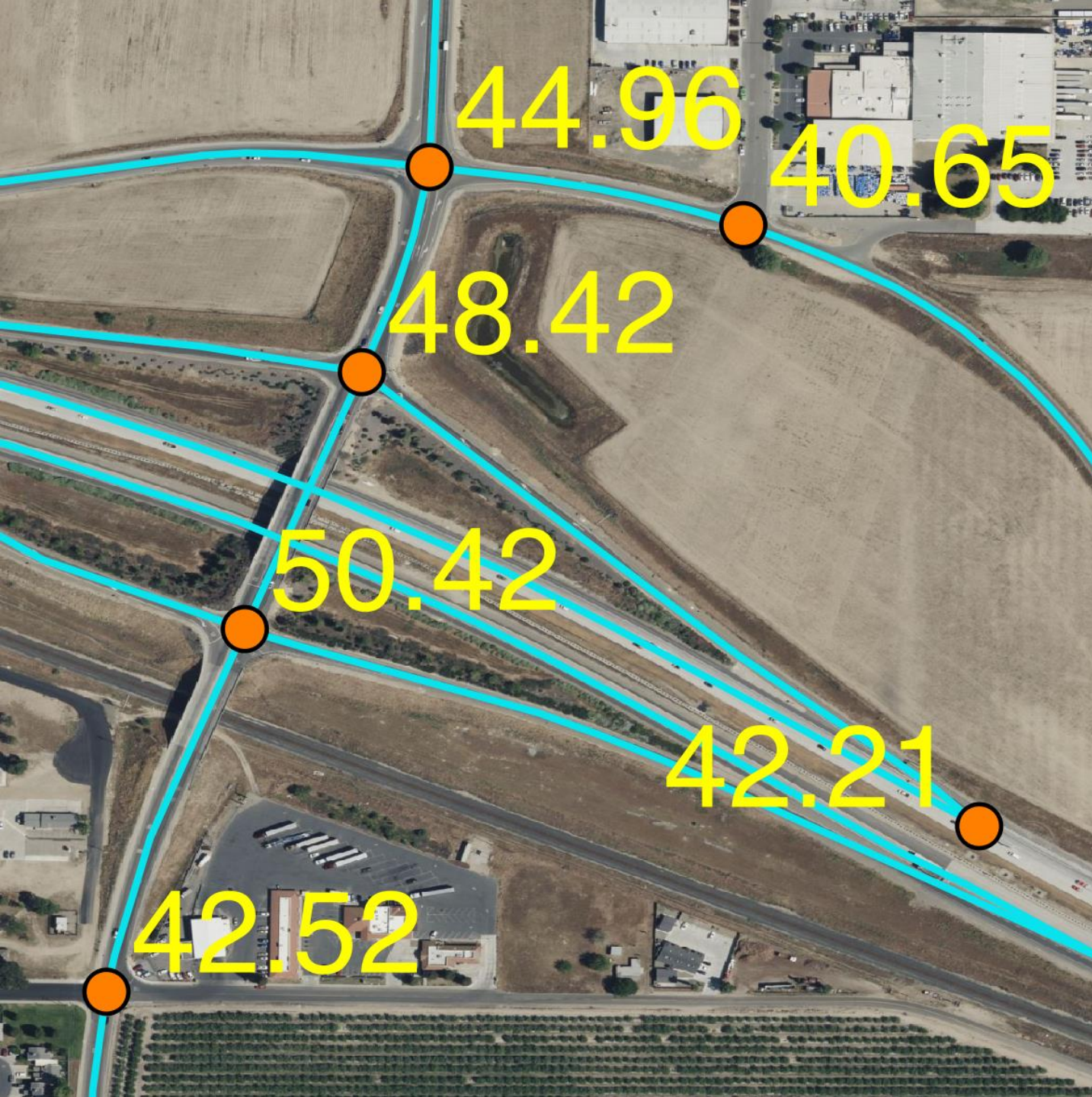
Network ID	1521
Closest city	Annaba
Country	Algeria
World Region	Middle East & North Africa
Elevated road structure	Bridge
Difference in elevation	3.22 meters





Network ID	1754
Closest city	Little Rock
Country	United States
World Region	North America
Elevated road structure	Overpass
Difference in elevation	3.37 meters





Network ID	1456
Closest city	Turlock
Country	United States
World Region	North America
Elevated road structure	Overpass
Difference in elevation	8.21 meters

